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ORIGINAL ARTICLE

Etiology, Management, and Outcomes of Society for Cardiovascular Angiography and Interventions Stage B Cardiogenic Shock

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BACKGROUND: There are limited data on the etiology, management, and outcomes of Society for Cardiovascular Angiography and Interventions (SCAI) B cardiogenic shock.

METHODS: From 2017 to 2022, adult patients (≥18 years) admitted to the medical, intermediate, and critical care units in a 6-hospital system were evaluated. SCAI B cardiogenic shock was defined as hypotension (systolic ≤90/mean ≤65 mm Hg) or hypoperfusion (lactate 2–5 mEq/L). Cardiac arrest, use of circulatory support, and noncardiac etiologies were excluded. The composite primary end point included transfer to a higher level of care, SCAI stage escalation, or in-hospital mortality. Multivariable analysis and mixed-effects regression models were used.

RESULTS: During this period, 500 patients (median age, 76 years; 56% men; 79% White) developed SCAI B cardiogenic shock (hypotension 18%, hypoperfusion 82%). The most common etiologies were heart failure (37%), arrhythmia (23%), and acute myocardial infarction (13%). The primary composite end point was noted in 135 patients (deterioration cohort). The deterioration cohort had comparable baseline characteristics to those who recovered, but before the primary outcome, had lower blood pressures, higher rates of renal (60% versus 33%) and hepatic (15% versus 4%) injury, less negative fluid balance (−0.30 versus −0.68 L), and greater diuretic resistance (21% versus 2%; $P<0.001$). In a multivariable analysis, acute kidney injury–adjusted odds ratio 2.17 (95% CI, 1.11–4.22); $P=0.02$–and diuretic resistance–adjusted odds ratio 9.55 (95% CI, 2.61–34.89); $P=0.001$–were independently predictive of clinical deterioration. Patients with isolated hypotension had worse outcomes compared with those with isolated hypoperfusion.

CONCLUSIONS: Among patients with SCAI B cardiogenic shock, a quarter of the population experienced clinical deterioration. Acute kidney injury and diuretic resistance in the preceding 24 hours were independently predictive of developing the primary end point.

Key Words: adult ■ blood pressure ■ critical care ■ heart failure ■ lactates

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Cardiogenic shock (CS) is characterized by cardiac dysfunction that presents as hypotension and hypoperfusion, leading to multisystem organ failure.[1,2] In efforts to standardize communication, triaging, and clinical severity, the Society for Cardiovascular Angiography and Interventions (SCAI) shock stage classification has been proposed to define CS severity based on vital signs, laboratory parameters, and treatment intensity.[3] Despite advances in drug and device technology, CS continues to be associated with high mortality (~50%), which may

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WHAT IS NEW?

- In early cardiogenic shock patients, a quarter of the population experienced clinical deterioration.
- Acute kidney injury and diuretic resistance in the preceding 24 hours were independently predictive of developing the primary end point.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Early recognition, detection of deterioration, and prompt management of early cardiogenic shock represent opportunities for improvement of outcomes in this population.

Nonstandard Abbreviations and Acronyms

AKI	acute kidney injury
AMI	acute myocardial infarction
CICU	cardiac intensive care unit
CS	cardiogenic shock
HF	heart failure
SCAI	Society for Cardiovascular Angiography and Intervention

be related to under-recognition of the disease state owing to differing presentations among CS etiologies.[4,5] Compared with acute myocardial infarction complicated by CS (AMI-CS), which has a clear onset of symptoms, patients with heart failure CS (HF-CS) may have a more indolent course characterized by hypoperfusion without hypotension.[6] In this preshock phase, corresponding to SCAI B CS, hypotension may occur as a late event, signaling collapse of compensatory neurohormonal and adrenergic mechanisms.[7,8] Prior data have demonstrated the importance of the Golden Day of CS, wherein the evolution of SCAI stages in the first 24 hours correlates with clinical outcomes.[9,10] If left untreated, a spiral of cardiometabolic dysfunction and multisystem organ failure is expected, thus underscoring the need for timely diagnosis and intervention.[1,11,12] The absence of hypotension has precluded characterization of a large portion of this vulnerable population due to delayed recognition.[13] Considering this information, we sought to evaluate SCAI B CS in a single health care system to allow for timely recognition and identification of predictors for clinical deterioration in this understudied cohort.

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METHODS

Study Setting

Brown University Health is a 6-hospital tertiary cardiovascular care system in Rhode Island and southern Massachusetts. Three hospitals have cardiac catheterization laboratories that can provide 24×7 full-spectrum percutaneous coronary interventions and percutaneous mechanical circulatory support therapies (intraaortic balloon pump, Impella CP [Danvers, MA]). The hub hospital has cardiac surgical, veno-arterial extracorporeal membrane oxygenation, and nonpercutaneous mechanical circulatory support (Impella 5.5, CentriMag) capabilities. Two of the 6 hospitals have established coronary care units that are able to provide specific cardiac critical care therapies, whereas the other 4 hospitals have mixed medical/surgical intensive care units. All hospitals are staffed by critical care medicine and cardiovascular disease physicians. The hospital network does not offer durable MCS or cardiac transplant but has a well-established pathway to a regional center (<100 miles distance) for such therapies.[5]

Patient Selection

From January 1, 2017 to December 31, 2022, we conducted a retrospective cohort study of all adults (≥18 years) admitted to the general medical, intermediate medical, respiratory, and cardiac step-down, cardiac intensive, and medical intensive care units with SCAI stage B CS. SCAI B CS was defined using a combination of the SCAI Shock staging and the Cardiogenic Shock Working Group modification of the SCAI Shock staging. Patients with isolated hypotension (systolic blood pressure ≤90 mm Hg or mean arterial pressure ≤65 mm Hg) for at least 2 consecutive vital sign checks or isolated hypoperfusion (serum lactate between 2 and 5 mEq/L) within the first 24 hours of admission were included.[3,14] Patients with a primary admission for severe sepsis, hypovolemia, dehydration, neurogenic shock, anaphylaxis, acute blood loss anemia, tension hydro- or pneumothorax, cardiac tamponade, compartment syndrome, hypothermia, orthostatic hypotension, adrenal insufficiency, decompensated cirrhosis, acute mesenteric ischemia, perforated viscus, interstitial lung disease, asthma, chronic obstructive pulmonary disease, seizure, alcohol withdrawal, ketoacidosis, and toxic ingestion were excluded.

The remaining cohort was reviewed by 2 independent reviewers to confirm the diagnosis of CS. A third reviewer adjudicated any discrepancies. Patients were included if they met the following admission or related diagnoses: congestive heart failure or cardiomyopathy, myocarditis, acute myocardial infarction, structural or valvular heart disease, or arrhythmia. Patients requiring vasoactive medications or mechanical circulatory support devices either temporarily or chronically were excluded. Patients who suffered any cardiac arrest or were placed on comfort measures/do-not-escalate care in the emergency department were similarly excluded. Detailed *International Classification of Diseases*, Tenth Revision, Clinical Modification coding criteria for the inclusion and exclusion diagnoses are detailed in Table S1. The study was approved by the Brown University Health institutional review board (2230674-2). The data, analytic methods, and study materials will not be made available to other researchers for purposes of reproducing the results or replicating the procedure.

Study Parameters, Definitions, and Outcomes

Clinical details from the index hospitalization included admission characteristics (method of triage, admission location, and hospital designation based on level of cardiac care), baseline demographics, comorbidities, admission vital signs, respiratory support, hemo-metabolic data, transthoracic echocardiography

parameters, invasive hemodynamics, and cardiac interventions. CS etiology was classified as AMI-CS (ST-segment–elevation MI or non–ST-segment–elevation MI with catheterization laboratory/angiographic details [Table S2]), HF-CS (de novo HF or acute-on-chronic HF), arrhythmia, structural or valvular heart disease, and miscellaneous (eg, infective endocarditis, fulminant myocarditis, and ≥1 CS etiology). Noncardiac organ dysfunction was detailed as AKI (Kidney Disease: Improving Global Outcomes criteria), acute liver injury (defined as alanine aminotransferase >200 IU/L), bacteremia (2 out of 2 blood cultures or 1 out of 2 for noncontaminants), acute hypoxic respiratory failure (arterial oxygen saturation <88% requiring supplemental oxygen), acidosis (defined as pH <7.30 of respiratory [$Paco_2$>50 mm Hg], metabolic [serum bicarbonate <22 mEq/L], or mixed components), and toxic metabolic encephalopathy).[1,15,16] Additional data were collected as a single measurement or in serial 8-hour increments 24 hours before the prespecified outcome (or before discharge for nondeteriorating patients) during the SCAI B CS course, including vital signs, respiratory support, diuretic regimen, urine output, fluid balance, and evidence of diuretic resistance (defined as urine output <0.5 cm^3/kg/h after a 1–1.5 mg/kg furosemide or equivalent diuretic challenge).[17,18] These data were obtained through chart review and manual extraction.

Outcomes

The primary composite outcome was defined as any of in-hospital mortality, intra- or interhospital transfer to higher levels of care, or escalation to a higher SCAI stage. Patients were evaluated as those who deteriorated (ie, ≥1 component of the primary outcome) or recovered (ie, discharge without any of the events) to identify predictors of clinical decline. Secondary end points of interest included hospital length of stay, discharge disposition, and 30-day hospital readmission rates in survivors.

Statistical Analysis

Univariate distributions of all variables are described. Continuous variables are presented as means±standard deviations or medians with interquartile ranges. Categorical variables are reported as frequencies and percentages. Comparisons of continuous variables were performed using the Student *t* test or Wilcoxon rank-sum test, where appropriate. Categorical variables were compared using the Fisher exact test. A $P<0.20$ in the univariate analysis satisfied inclusion in the multivariable logistic regression model for predictors of deterioration. Outcomes are presented as adjusted odds ratios with 95% CIs. Model fit was assessed using the Hosmer-Lemeshow goodness-of-fit test. Mixed-effects regression models were used to assess differences over time in serial vital signs between the 2 groups from admission to time of outcome. The random effects were specified as repeated and longitudinal time points whereas the fixed effects referred to deterioration. Time to the first component of the composite outcome within isolated hypotension and isolated hypoperfusion subgroups was assessed using Kaplan-Meier plots. Differences in the curves were compared using the log-rank test. A Cox proportional-hazards model was used to estimate the hazard ratio. All tests were 2-sided, and $P\leq0.05$ was considered statistically significant. All analyses were performed using Stata/MP 18.0 (College Station, TX).

RESULTS

Baseline Characteristics

In this 6-year period, 500 patients with SCAI stage B CS met our inclusion criteria—84 (17%) with isolated hypotension and 416 (83%) with isolated hypoperfusion, of whom 135 patients deteriorated (27%). The mean age was 76 years, 56% were men, and 80% were of White race (Table 1). Heart failure, arrhythmia, and acute myocardial infarction were the predominant etiologies in this population (Figure 1), with no statistically significant differences between those who deteriorated and those who recovered (Table 1). Over half the patients (54%) were admitted to the medical floors, with the remaining admitted to intermediate care (10%) and cardiac or medical intensive care units (37%). Only 24% of the patients had the term shock documented in the electronic medical record during the admission, with higher rates in those who deteriorated (54% versus 13%; $P<0.001$).

Shock Characteristics and Interventions

Compared with those who recovered, patients who deteriorated had higher predeterioration hypotension, B-type natriuretic peptide, serum creatinine, AKI, acute liver injury, toxic metabolic encephalopathy, oliguria, and lower left ventricular ejection fraction, tricuspid annular plane systolic excursion ratio, and urine output (Table 2). Among those with available invasive hemodynamic data (9%), those who deteriorated also had higher right-sided pressures, and lower cardiac output and cardiac power output. The 2 cohorts had largely comparable rates of in-hospital interventions, however, the deterioration cohort had poorer decongestion efficiency (ie, lower urine output with greater diuretic resistance) despite higher loop diuretic dosing (Table 2).

There were no statistically significant differences between the 2 groups pertaining to maximum respiratory support or continuous infusions of nitroglycerin or antiarrhythmic therapies within the first 24 hours of admission. However, the deterioration cohort had a lower urine output, a less net negative fluid balance, and more oliguria despite equivalent diuretic regimen intensity over the first 24 hours. The deterioration cohort had greater duration in SCAI stage B CS—11 (interquartile range, 5–27) versus 5 (interquartile range, 2–15) hours; $P<0.001$ (Table S3).

Serial Vital Signs

Serial vital signs were recorded in real time and obtained through chart review in patients who recovered or deteriorated during the 24 hours before the clinical outcome. The deterioration cohort had consistently lower mean arterial pressures, higher respiratory rates, and greater use of noninvasive positive pressure ventilation in the

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Table 1. Baseline Characteristics of SCAI B Cardiogenic Shock Cohort

Characteristic	Total cohort (N=500)	Recovery cohort (n=365)	Deterioration cohort (n=135)	*P*
Demographics				
Age, y	76 (68–85)	77 (68–85)	74 (67–84)	0.34
Male sex	281 (56.2)	202 (55.3)	79 (58.5)	0.54
Race				0.09
White	398 (79.6)	288 (78.9)	110 (81.5)	
Black	46 (9.2)	40 (10.9)	6 (4.4)	
Asian	7 (1.4)	5 (1.4)	2 (1.5)	
Hispanic/Latino	45 (9.0)	30 (8.2)	15 (11.1)	
Other	4 (0.80)	2 (0.55)	2 (1.5)	
Body mass index, kg/m^2	27.0 (22.7–32.2)	26.8 (22.7–32.0)	27.9 (22.9–33.3)	0.33
Comorbidities				
Hypertension	414 (82.8)	305 (83.6)	109 (80.7)	0.51
Hyperlipidemia	319 (63.8)	237 (64.9)	82 (60.7)	0.40
Diabetes	205 (41.1)	150 (41.1)	55 (41.0)	>0.99
Congestive heart failure	258 (51.6)	190 (52.1)	68 (50.4)	0.76
Atrial fibrillation or flutter	230 (46.0)	175 (47.9)	55 (40.7)	0.16
Obstructive sleep apnea	50 (10)	37 (10.1)	13 (9.6)	>0.99
Chronic lung disease	126 (25.2)	87 (23.8)	39 (28.9)	0.25
Chronic kidney disease*				
Stage II	15 (3.0)	10 (2.7)	5 (3.7)	0.56
Stage IIIa	70 (14.0)	50 (13.7)	20 (14.8)	0.77
Stage IIIb	38 (7.6)	23 (6.3)	15 (11.1)	0.09
Stage IV	25 (5.0)	17 (4.7)	8 (5.9)	0.64
Stage V	26 (5.2)	17 (4.7)	9 (6.7)	0.37
Coronary artery disease	256 (51.2)	187 (51.2)	69 (51.1)	>0.99
Peripheral artery disease	66 (13.2)	50 (13.7)	16 (11.9)	0.66
Prior ICD or PPM	62 (12.4)	44 (12.1)	18 (13.3)	0.76
Prior PCI or CABG	135 (27.0)	93 (25.5)	42 (31.1)	0.21
Admission characteristics				
Method of inpatient admission				>0.99
Emergency department	485 (97)	360 (98.6)	130 (96.3)	
Direct or scheduled admission	5 (1)	4 (1.1)	…	
Outside hospital transfer	10 (2)	1 (0.3)	5 (3.7)	
Triage unit				
Floors	268 (53.6)	203 (55.6)	65 (48.1)	0.16
Intermediate care	49 (9.8)	34 (9.3)	15 (11.1)	0.61
Critical care	183 (36.6)	128 (35.1)	55 (40.7)	0.25
Hospital level of cardiac care				
Level I	318 (63.6)	234 (64.1)	84 (62.2)	0.75
Level II	170 (34)	122 (33.4)	48 (35.6)	0.67
Level III	11 (2.2)	9 (2.5)	2 (1.5)	0.74
Etiology				
Acute myocardial infarction	66 (13.2)	47 (12.9)	19 (14.1)	>0.99
ST-segment–elevation	11 (16.7)	8 (17.0)	3 (15.8)	
Heart failure	184 (36.8)	131 (35.9)	53 (39.3)	0.53
Acute-on-chronic heart failure	147 (79.9)	108 (82.4)	39 (73.6)	
De novo heart failure	29 (15.8)	18 (13.7)	11 (20.8)	

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Table 1. Continued

Characteristic	Total cohort (N=500)	Recovery cohort (n=365)	Deterioration cohort (n=135)	*P*
Isolated right ventricular failure	7 (4.3)	5 (3.9)	2 (5.6)	
Arrhythmia	117 (23.4)	93 (25.5)	24 (17.8)	0.08
Structural or valvular heart disease	16 (3.2)	10 (2.7)	6 (4.4)	0.39
Miscellaneous†	117 (23.4)	84 (23)	33 (24.4)	>0.99

Median (interquartile range) or number (percentage). CABG indicates coronary artery bypass grafting; ICD, implantable cardioverter defibrillator; PCI, percutaneous coronary interventions; PPM, permanent pacemaker; and SCAI, Society for Cardiovascular Angiography and Intervention.

*Stages are separate variables, does not sum to 100%.

†Composite of infective endocarditis, fulminant myocarditis, and shock due ≥1 etiology (view Figure 1 for regions of overlap).

preceding 24 hours of the primary composite outcome (Figure 2).

Clinical Outcomes

The composite primary outcome of in-hospital mortality, escalation of SCAI stage, or escalation to a higher level of care occurred in 135 (27%) patients. Among these, in-hospital mortality was observed in 52 (39%) patients, progression of CS stage in 81 (60%), and escalation to a higher level of care in 66 (49%) (Table 3). The deterioration cohort had longer hospitalization duration, fewer discharges to home, more discharges to hospice, and higher rates of transfer to durable MCS/cardiac transplantation centers. Both groups had comparable rates of 30-day hospital readmission, with most cardiac-related causes attributable to heart failure (Table 3), and no differences in long-term mortality among survivors (Figure S1). In a multivariable analysis, the presence of acute kidney injury (AKI)—adjusted odds ratio 2.17 (95% CI, 1.11–4.22); *P*=0.021 and diuretic resistance during the 24 hours before the outcome—adjusted odds ratio 9.55 (95% CI, 2.61–34.89); *P*=0.001 were noted to be predictors of the primary composite end point (Figure 3 and Table S4).

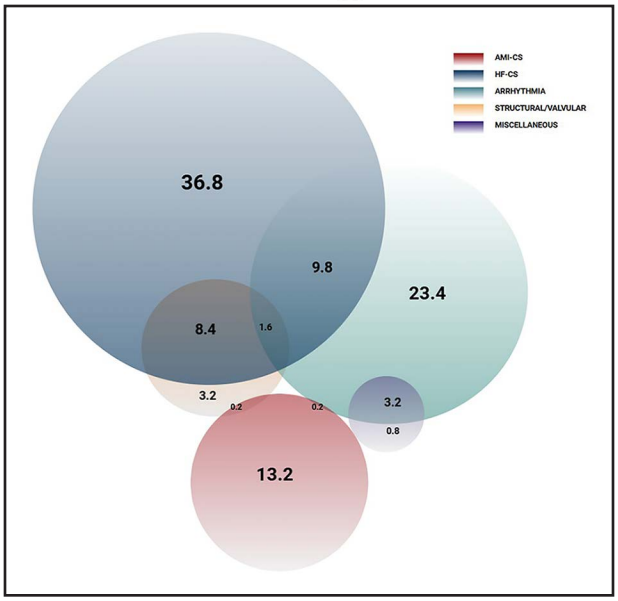

Figure 1. Etiology of Society for Cardiovascular Angiography and Intervention (SCAI) stage B cardiogenic shock (CS). AMI indicates acute myocardial infarction; and HF, heart failure.

Isolated Hypotension Versus Isolated Hypoperfusion

In a prespecified subgroup analysis, we evaluated patients with isolated hypotension (91% to 18%) versus isolated hypoperfusion (409% to 82%). Compared with isolated hypoperfusion, those with isolated hypotension had higher rates of the individual components and the primary composite outcome—hazard ratio, 2.69 (95% CI, 1.89–3.83); log-rank test *P*<0.001 (Table 4 and Figure 4).

DISCUSSION

In this retrospective single hospital system analysis of SCAI stage B CS, the composite primary outcome of in-hospital death, escalation of SCAI stage, or transfer to a higher level of care was observed in over a quarter of the patients. The main findings were that (1) patients with isolated hypotension experienced worse outcomes compared with those with isolated hypoperfusion, and (2) the independent predictors of deterioration were AKI and diuretic resistance.

Prior studies of CS have largely focused on SCAI stages C–E,[19,20] with limited data available on stage B.[8,13] Data from the Critical Care Cardiology Trials Network across 47 cardiac intensive care units (CICUs) evaluated preshock cohorts—defined as isolated low cardiac output, isolated hypotension, and normotensive CS—and compared them with classical patients with CS (8). Among 597 preshock patients, isolated hypoperfusion was associated with higher in-hospital mortality than isolated hypotension (17% versus 11%), with 30% of hypotensive patients eventually developing hypoperfusion.[8] A single-center CICU study similarly demonstrated mortality rates of 9% and 17% for isolated hypotension and isolated hypoperfusion, respectively.[13] Importantly, both studies focused on dedicated CICU populations, with the former conducted after publication of the SCAI classification. In contrast, the current study deliberately

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Table 2. SCAI Stage B Cardiogenic Shock Characterization and Interventions

Characteristics	Total cohort (N=500)	Recovery cohort (n=365)	Deterioration cohort (n=135)	*P*
Admission vitals				
Heart rate, beats per minute	100 (81–122)	101 (80–122)	98 (82–122)	0.63
Systolic blood pressure, mm Hg	125 (102–149)	130 (107–150)	108 (88–137)	<0.001
Diastolic blood pressure, mm Hg	75 (59–89)	77 (61–91)	66 (55–85)	<0.001
Mean arterial pressure, mm Hg	93 (75.3–108)	96.3 (81–110.3)	83 (67–100)	<0.001
Pulse pressure, mm Hg	48 (35–63)	51 (38–65)	41 (32–56)	<0.001
Respiratory rate, per min	20 (18–26)	20 (18–26)	20 (18–28)	0.66
Temperature, °F	97.6 (97.0–98.2)	97.6 (97.1–98.2)	97.4 (96.9–98.2)	0.13
Laboratory parameters				
B, type natriuretic peptide, pg/mL	834 (416–1687)	760 (392–1617)	970 (506–1838)	0.042
High-sensitivity Troponin-I, ng/L	79 (32–283)	76 (29–258)	102 (38–537)	0.12
Venous pH	7.33 (7.27–7.39)	7.34 (7.27–7.40)	7.32 (7.27–7.38)	0.14
Lactate, mEq/L	2.5 (2.1–3.3)	2.5 (2.2–3.2)	2.6 (2.0–3.5)	0.81
Creatinine, mg/dL	1.22 (0.93–1.78)	1.17 (0.92–1.53)	1.59 (0.98–2.25)	<0.001
eGFR, mL/min per 1.73 m^2	49 (32–60)	53 (38–60)	39 (25–60)	<0.001
Alanine transaminase, IU/L	19 (12–39)	19 (12–36)	23 (12–58)	0.28
Hemoglobin, g/dL	12.2 (10.2–13.9)	12.2 (10.2–14)	12.1 (10.1–13.9)	>0.99
Transthoracic echocardiography				
Left ventricular ejection fraction, %	40 (25–56)	42 (30–60)	35 (21–55)	0.08
TAPSE, mm	16 (12–20)	17 (13–21)	14.5 (11–18)	<0.001
RVSP, mm Hg	42 (35–54)	40 (33–52)	46 (38–57)	0.009
Inferior vena caval size, mm	20.5 (17–24)	20 (16–24)	22 (19–25)	0.14
Invasive hemodynamics				
Right heart catheterization	47 (9.4)	18 (4.9)	29 (21.5)	<0.001
Right atrial pressure, mm Hg	11 (7–15)	9 (7–12)	12 (8–18)	0.07
Mean PAP, mm Hg	34 (27–45)	30 (22–35)	40 (28–48)	0.012
Mixed venous oxygen saturation	57 (48–66)	63 (49–69)	53 (47–59)	0.031
Pulmonary artery pulsatility index	2.49 (1.9–3.3)	2.5 (2.1–3.5)	2.46 (1.8–3.6)	0.70
Pulmonary vascular resistance, WU	2.7 (1.4–4.7)	2 (1.3–2.8)	3.1 (1.7–4.9)	0.031
PCWP, mm Hg	22 (15–30)	20 (13–22)	25 (17–30)	0.06
Cardiac output, L/min; Fick	3.6 (3.1–5.3)	3.9 (3.1–5.6)	3.3 (3–4.1)	0.040
Cardiac index, L/min/m^2; Fick	2 (1.7–2.7)	2.2 (2–2.9)	1.9 (1.5–2.3)	0.042
Cardiac power output, W	0.63 (0.53–0.95)	0.77 (0.55–1.4)	0.61 (0.48–0.76)	0.026
SVR, dynes/s per cm^{-5}	1642 (1250–1977)	1414 (1196–1777)	1879 (1320–2074)	0.16
Noncardiac organ dysfunction				
Acute kidney injury	203 (40.6)	122 (33.4)	81 (60)	<0.001
KDIGO stage				<0.001
KDIGO I	125 (61.6)	94 (77.1)	31 (38.3)	
KDIGO II	44 (21.7)	20 (16.4)	24 (29.6)	
KDIGO III	34 (16.8)	8 (6.6)	26 (32.1)	
Acute liver injury	33 (6.6)	13 (3.6)	20 (14.8)	<0.001
Acute hypoxic respiratory failure	379 (75.8)	273 (74.8)	106 (78.5)	0.41
Toxic metabolic encephalopathy	53 (10.6)	28 (7.7)	25 (18.5)	<0.001
Acidosis	147 (29.4)	104 (28.5)	43 (31.9)	0.51
Respiratory	111 (75.5)	74 (71.2)	37 (86)	
Metabolic	11 (7.5)	9 (8.6)	2 (4.7)	
Mixed	25 (17)	21 (20.2)	4 (9.3)	

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Table 2. Continued

Characteristics	Total cohort (N=500)	Recovery cohort (n=365)	Deterioration cohort (n=135)	*P*
Bronchopneumonia	66 (13.2)	47 (12.9)	19 (14.1)	0.77
Bacteremia	16 (3.2)	8 (2.2)	8 (5.9)	0.052
Decongestion profile 24 h before outcome				
Urine output, L	1.1 (0.55–1.9)	1.3 (0.6–2.0)	0.96 (0.3–1.7)	0.001
Net negative balance, L	0.60 (1.30–0)	0.68 (1.40–0.12)	0.30 (1.10–0.10)	0.012
Diuretic resistance	35 (7.0)	7 (1.9)	28 (20.9)	<0.001
Oliguria over last 24 h	146 (40.6)	98 (37.6)	48 (48.5)	0.07
Intravenous furosemide equivalents, mg	20 (0, 80)	20 (0–40)	40 (0–80)	0.024

Median (interquartile range) or number (percentage). eGFR indicates estimated glomerular filtration rate; KDIGO, Kidney Disease: Improving Global Outcomes; PAP, pulmonary artery pressure; PAPi, pulmonary artery pulsatility index; PCWP, pulmonary capillary wedge pressure; RV, right ventricle; RVSP, right ventricular systolic pressure; SCAI, Society for Cardiovascular Angiography and Interventions; STEMI, ST, segment, elevation myocardial infarction; SVR, systemic vascular resistance; and TAPSE, tricuspid annular plane systolic excursion ratio.

examined an all-comer population not restricted to the CICU. This distinction is meaningful: recognition of CS may be more intuitive in a CICU than in noncardiac ICUs or general wards, and monitoring frequency for hypotension is substantially lower outside the ICU. Furthermore, unlike the Critical Care Cardiology Trials Network study—where pulmonary artery catheterization contributed to classification—our study relied on routinely available clinical variables, aligning with the low rates of invasive monitoring typically seen in early shock.[21,22] Pulmonary artery catheterization was used in only 9% of our population, similar to rates in the Mayo Clinic CICU cohort.[13]

A key divergence between the original SCAI framework and the SCAI-CSWG modification lies in the definition of SCAI Stage B (preshock). The original classification designates Stage B as relative or absolute hypotension without any biochemical or clinical evidence of hypoperfusion. In contrast, the SCAI-CSWG criteria allow stage B CS designation for patients with either absolute hypotension or early hypoperfusion (lactate 2–5 mmol/L). Clarifying this discrepancy is clinically significant, as ≈1 in 4 patients meeting SCAI-CSWG stage B criteria in this work experience adverse outcomes and demonstrate the highest likelihood of progressing to a higher shock stage within 24 hours. However, unlikely prior data that demonstrated patients with isolated hypoperfusion to have worse outcomes that those with hypotension, we noted the converse.[13] A plausible mechanistic explanation for our findings is that patients with isolated hypoperfusion (ie, normotensive CS) may still benefit from preserved compensatory mechanisms that maintain blood pressure despite early tissue oxygen debt. In contrast, hypotension in the absence of overt hypoperfusion may signal a more advanced or rapidly evolving loss of

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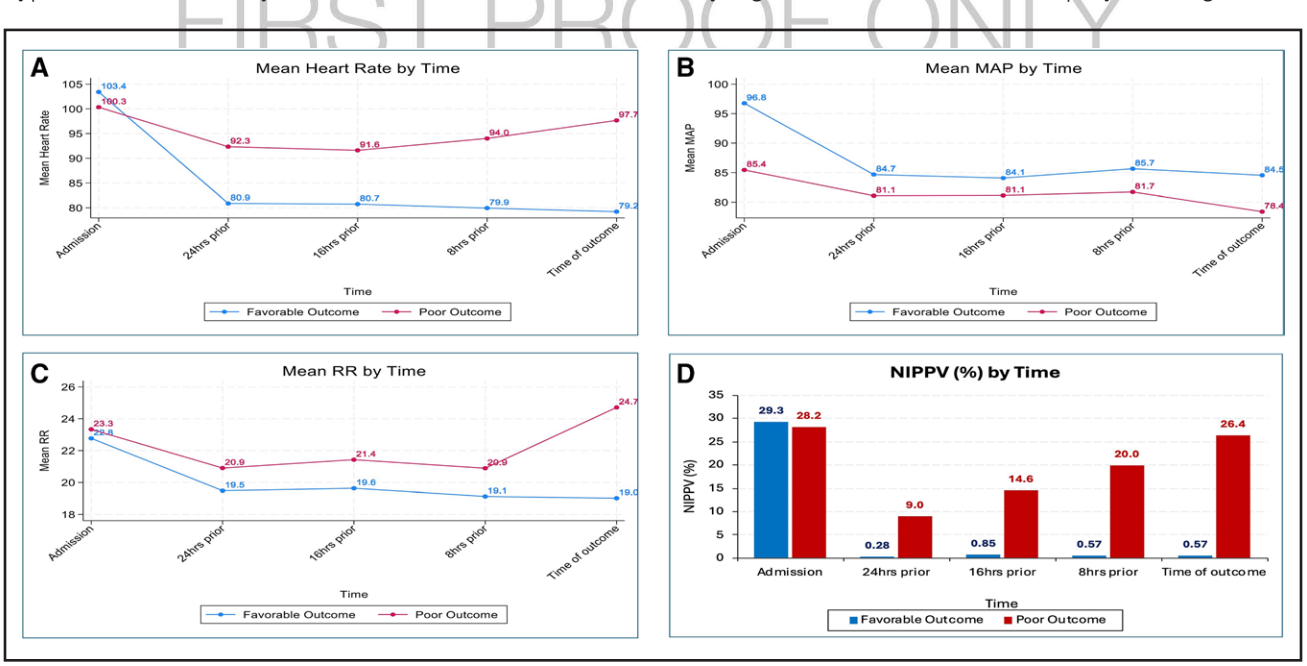

Figure 2. Trends in vital signs and respiratory support before outcome.
Serial 8-hour comparisons in heart rate (**A**), mean arterial pressure (MAP) (**B**), respiratory rate (**C**), and frequency of noninvasive positive pressure ventilation (NIPPV) (**D**) between favorable and poor outcome cohorts in the prior 24 hours. RR indicates respiratory rate.

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Table 3. Depiction of Individual Components of Primary Composite Outcome and Secondary End Points

In, hospital outcomes	Total cohort (N=500)	Recovery cohort (n=365)	Deterioration cohort (n=135)	*P*
In-hospital mortality	52 (10.4)	…	52 (38.5)	…
SCAI stage escalation				…
B	418 (83.8)	365 (100)	53 (39.6)	
C	40 (8.0)	…	40 (29.9)	
D	30 (6.0)	…	30 (22.4)	
E	11 (2.2)	…	11 (8.2)	
Transferred to higher level care				…
No transfer	434 (86.8)	365 (100)	69 (51.1)	
Intermediate cardiac care	10 (2.0)	…	10 (7.4)	
Coronary care unit	34 (6.8)	…	34 (25.2)	
Medical intensive care unit	19 (3.8)	…	19 (14.1)	
Interhospital	3 (0.60)	…	3 (2.2)	
Discharge disposition				<0.001
Home ± services	262 (52.7)	224 (61.4)	38 (28.8)	
SNF or rehabilitation	134 (26.9)	109 (29.9)	25 (18.9)	
Hospice	39 (7.9)	24 (6.6)	14 (10.6)	
Against medical advice	6 (1.2)	4 (1.1)	2 (1.5)	
Transfer LVAD/HTx center	9 (1.8)	2 (0.55)	7 (5.3)	
Post, hospitalization outcomes	**Total cohort (N=406)**	**Recovery cohort (n=333)**	**Deterioration cohort (n=73)**	***P***
30 d hospital readmission	84 (18.8)	71 (19.5)	13 (15.6)	0.52
Cardiac, related causes	47 (56)	39 (55)	8 (62)	
Heart failure	31 (36.9)	24 (33.9)	7 (53.8)	
Myocardial infarction	2 (2.4)	3 (4.2)	…	
Others*	14 (16.7)	12 (16.9)	2 (8.2)	
Noncardiac-related causes	37 (44)	32 (45)	5 (38)	
30-day all-cause mortality	40 (9.9)	30 (9)	10 (13.7)	0.28
1-year all-cause mortality	144 (35.2)	121 (36.3)	23 (31.5)	0.38

Number (percentage). HTx, heart transplantation; LVAD, left ventricular assist device; SCAI, Society for Cardiovascular Angiography and Interventions; SNF, skilled nursing facilities.

*Includes arrhythmias, structural or valvular heart disease, miscellaneous, and ≥1 etiology.

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cardiovascular reserve, even before biochemical markers of organ dysfunction seem. Notably, most patients in our cohort had acute-on-chronic HF-CS, a state characterized by standing reliance on elevated sympathetic tone and other compensatory pathways. In such patients, the development of hypotension likely represents the exhaustion of these chronic compensatory mechanisms and therefore serves as a late and high-risk marker of more advanced shock physiology.[6]

This study also highlights delays in prompt CS diagnosis. Despite vital sign changes suggesting impending decompensation (ie, higher rates of tachycardia, hypotension, tachypnea, and use of respiratory support), a longer duration in SCAI stage B CS (11 versus 5 hours), and globally compromised renal parameters in the preceding 24 hours, only half of the deterioration group received the diagnosis of shock in the medical record during their hospitalization. These data are consistent with similar time-sensitive critical care syndromes.[23] The majority of the study population were admitted to noncardiac units and met SCAI stage B criteria via hypoperfusion and not hypotension (ie, normotensive CS), which likely accounts for delays in shock recognition. While prior data have highlighted the vitality of CS teams in CS care,[24] the challenge in early CS remains prompt recognition and escalation of care. Outside of hypotension, which can be assessed at frequent intervals in most patients, all other parameters of CS assessment require a high index of suspicion and frequent reassessment. This may often result in delayed diagnosis and management contributing to worse outcomes.[25] The development of a convenient, reproducible EMR-based CS detection software that provides clinical and research support can potentially aid in the early recognition, optimal treatment, and improvement in clinical outcomes.[25,26]

Last, this study provides novel insights into the cardiorenal cascade in SCAI stage B CS. While the role of AKI has been well established in higher stages of CS,[27,28] the role of cardiorenal dysfunction is less understood in

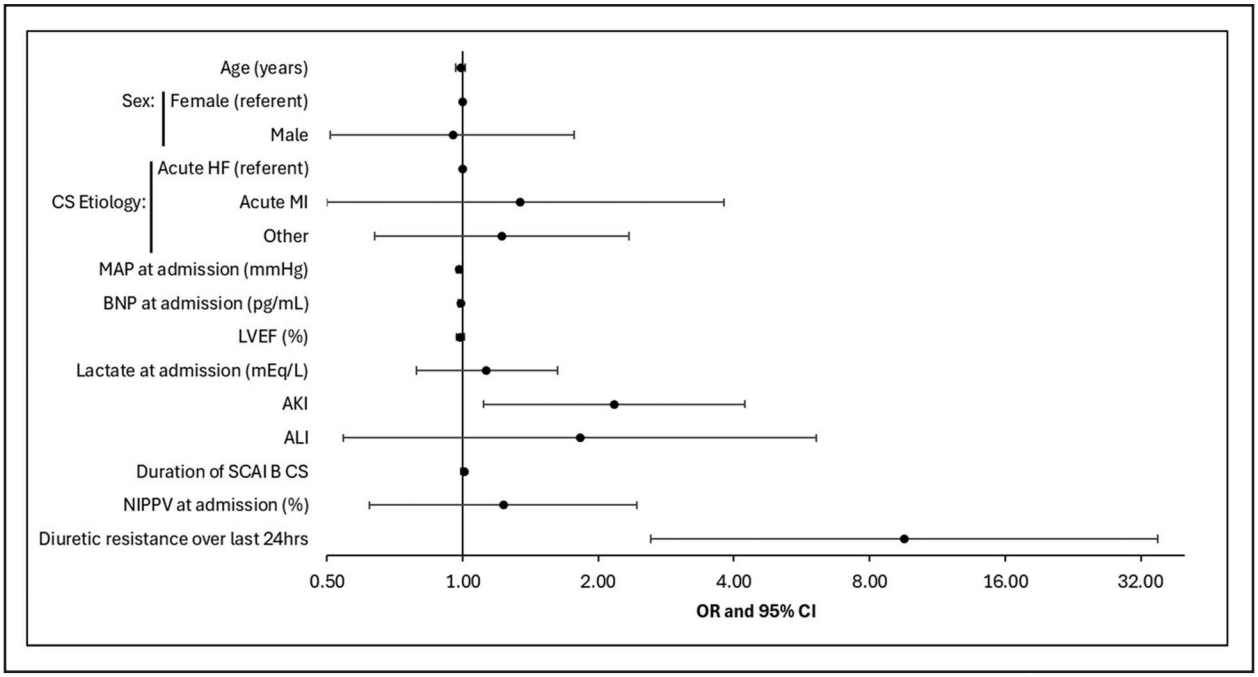

Figure 3. Multivariable analysis of predictors of poor outcomes in Society for Cardiovascular Angiography and Intervention (SCAI) stage B cardiogenic shock (CS).
Forrest plot depicting independent clinical predictors for the primary composite outcome of SCAI stage escalation, transfer to higher level of care, or inpatient mortality. AKI indicates acute kidney injury; ALI, acute liver injury; BNP, B-type natriuretic peptide; HF, heart failure; LVEF, left ventricular ejection fraction; MAP, mean arterial pressure; MI, myocardial infarction; and NIPPV, noninvasive positive pressure ventilation.

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early CS.[29,30] Prior work from the CSWG has demonstrated that worsening renal failure, as measured by an increase in serum creatinine, decrease in estimated glomerular filtration rate, or initiation of renal replacement therapy, was associated with higher in-hospital mortality.[20] This current study provides more granular information relating to urine output, fluid balance, diuretic dosing, detailed AKI staging, and diuretic resistance, all of which provide real time actionable information for influencing the trajectory of CS. AKI in the setting of CS is multifold and involves both hypoperfusion and renal venous congestion.[16,30] In this work, patients received suboptimal diuretic dosing to achieve an adequate 1 to 1.5 mg/kg furosemide challenge which may have contributed to worsening congestion and progression toward diuretic resistance.[12,17,18] Diuretic resistance therefore can serve as a physiological marker of severe renal venous congestion with threatened perfusion

Table 4. Primary Composite and Ancillary Outcomes Stratified by SCAI B CS Criteria

Outcomes	Total cohort (N=500)	Isolated hypotension (n=91)	Isolated hypoperfusion (n=409)	*P*
Poor outcome	135 (27.0)	44 (48.4)	91 (22.3)	<0.001
In, hospital mortality	52 (10.4)	14 (15.4)	38 (9.3)	0.09
SCAI stage escalation				<0.001
B	418 (83.6)	58 (63.7)	360 (88.2)	
C	40 (8)	16 (17.6)	24 (5.9)	
D	30 (6)	12 (13.2)	18 (4.4)	
E	11 (2.4)	5 (5.5)	7 (1.7)	
Transfer to higher level care				0.001
No transfer	434 (86.8)	68 (74.7)	366 (89.5)	
Intermediate care	10 (2)	2 (2.2)	8 (1.9)	
Critical care	53 (10.6)	19 (20.9)	34 (8.3)	
Inter, hospital	3 (0.6)	2 (2.2)	1 (0.3)	
Hospital length of stay, d	6 (3–9)	6 (4–10)	5 (3–9)	0.22
Longest duration in stage B, h	6 (2–19)	5 (2–16)	7 (3–19)	0.27

Median (interquartile range) or number (percentage). SCAI indicates Society for Cardiovascular Angiography and Interventions.

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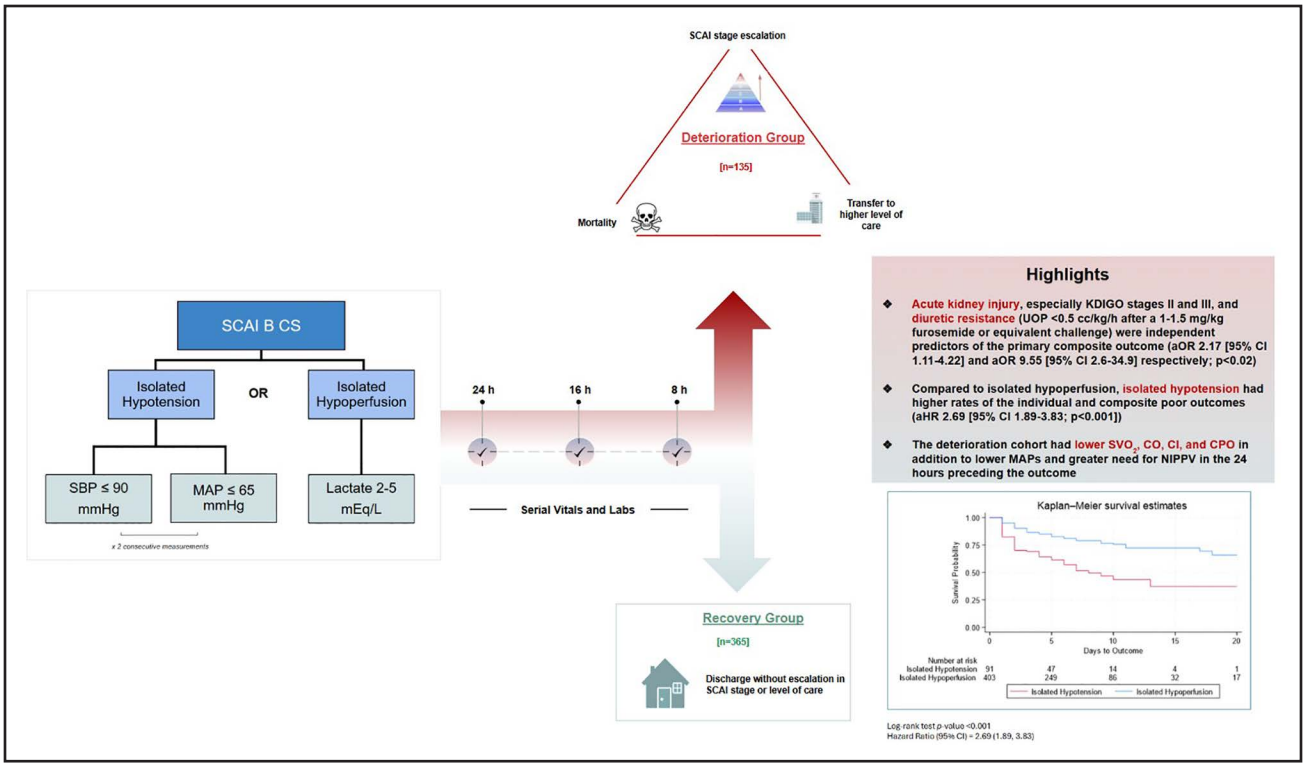

Figure 4. Etiology, management, and outcomes of Society for Cardiovascular Angiography and Intervention (SCAI) stage B cardiogenic shock (CS).
aOR indicates adjusted odds ratio; aHR, adjusted hazard ratio; CI, cardiac index; CPO, cardiac power output; CO, cardiac output; CS, cardiogenic shock; HR, hazard ratio; KDIGO, Kidney Disease: Improving Global Outcomes; MAP, mean arterial pressure; NIPPV, noninvasive positive pressure ventilation; SBP, systolic blood pressure; SVO2, venous oxygen saturation; and UOP, urine output.

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to support earlier care escalation before overt shock develops.

Limitations

This study has limitations, some of which are inherent to a retrospective, single health care system design subject to local practice patterns. During the study duration, Brown University Health did not have a unified shock team or protocol to standardize CS management. Further the electronic medical record was consolidated to a uniform output in 2017, and therefore, data before then was not collected. Second, the time frame of the study overlaps with the SARS-CoV-2 pandemic which likely affected resource allocation and intensity of care for SCAI B patients, given the majority were normotensive. The SCAI classification was also not available during the initial duration of this study, which may have contributed to under-recognition of the preshock state Third is that not all CS requires diuresis. Acute right ventricular failure requires preload to augment transpulmonary flow. Similarly, small decreases in preload can lead to cardiovascular collapse in patients with severe aortic stenosis. Similarly, small decreases in preload can lead to cardiovascular collapse in patients with severe aortic stenosis or dynamic outflow tract obstruction. Our study did not measure serial lactates in the population with initial hypoperfusion. In recent times, there has been increasing interest in trending serial lactate values to assess resolution of CS and represents a logical next step for future investigation in the SCAI B population.[31] Last, our study did not classify patients as SCAI stage B on the basis of tachycardia (ie, heart rate >100 beats per minute) as a previously suggested criterion for relative hypotension,[3] given the difficulty of retrospectively attributing isolated tachycardia to a cardiac etiology. Moreover, while tachycardia may serve as an early marker of impending deterioration in AMI- and HF-related CS, reliance on this criterion alone disadvantages CS due to bradyarrhythmias.

Conclusions

In this large multiyear study from a 6-hospital health care system, over a quarter of the patients with SCAI stage B CS suffered the primary composite outcome of in-hospital mortality, escalation of SCAI stage, or transfer to a higher level of care. AKI and diuretic resistance were predictive of poor outcomes. Further dedicated research, including use of electronic medical records-based detection techniques, for early diagnosis and evaluation of

SCAI B CS is urgently required to improve clinical outcomes in this population.

ARTICLE INFORMATION

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Supplemental Material

Table S1–S4
Figure S1

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REFERENCES

1. Vallabhajosyula S, Dunlay SM, Prasad A, Kashani K, Sakhuja A, Gersh BJ, Jaffe AS, Holmes DR Jr, Barsness GW. Acute noncardiac organ failure in acute myocardial infarction with cardiogenic shock. *J Am Coll Cardiol*. 2019;73:1781–1791. doi: 10.1016/j.jacc.2019.01.053
2. Vallabhajosyula S, Nagaraja V. Risk prediction and standardisation of cardiogenic shock care. *Heart Lung Circ*. 2024;33:403–405. doi: 10.1016/j.hlc.2024.04.001
3. Naidu SS, Baran DA, Jentzer JC, Hollenberg SM, van Diepen S, Basir MB, Grines CL, Diercks DB, Hall S, Kapur NK, et al. SCAI SHOCK stage classification expert consensus update: a review and incorporation of validation studies: this statement was endorsed by the American College of Cardiology (ACC), American College of Emergency Physicians (ACEP), American Heart Association (AHA), European Society of Cardiology (ESC) Association for Acute Cardiovascular Care (ACVC), International Society for Heart and Lung Transplantation (ISHLT), Society of Critical Care Medicine (SCCM), and Society of Thoracic Surgeons (STS) in December 2021. *J Am Coll Cardiol*. 2022;79:933–946. doi: 10.1016/j.jacc.2022.01.018
4. Bansal K, Gupta M, Garg M, Patel N, Truesdell AG, Babar Basir M, Rab ST, Ahmad T, Kapur NK, Desai N, et al. Impact of inpatient percutaneous coronary intervention volume on 30-day readmissions after acute myocardial infarction-cardiogenic shock. *JACC Heart Fail*. 2024;12:2087–2097. doi: 10.1016/j.jchf.2024.07.014
5. Mehta C, Has P, Osorio B, Godding M, Khetpal V, Clancy A, Mehta A, Poppas A, Kataria R, Saad M, et al. Serial severity assessment in advanced stages of AMI-cardiogenic shock requiring micro-axial flow pump support. *J Soc Cardiovasc Angiogr Interv*. 2025;4:102565. doi: 10.1016/j.jscai.2025.102565
6. Abraham J, Blumer V, Burkhoff D, Pahuja M, Sinha SS, Rosner C, Vorovich E, Grafton G, Bagnola A, Hernandez-Montfort JA, et al. Heart failure-related cardiogenic shock: pathophysiology, evaluation and management considerations: review of heart failure-related cardiogenic shock. *J Card Fail*. 2021;27:1126–1140. doi: 10.1016/j.cardfail.2021.08.010
7. Mehta A, Bansal M, Belford PM, Gilbert ON, Zhao DX, Vallabhajosyula S. Society of Cardiovascular Angiography and intervention Stage-B cardiogenic shock: an interventional-heart failure-critical care conundrum. *IHJ Cardiovasc Case Rep*. 2023;7:76–80. doi: 10.1016/j.ihjccr.2023.07.002
8. Patel SM, Berg DD, Bohula EA, Baird-Zars VM, Park JG, Barnett CF, Daniels LB, Fordyce CB, Ghafghazi S, Goldfarb MJ, et al. Continuum of preshock to classic cardiogenic shock in the Critical Care Cardiology Trials Network Registry. *JACC Heart Fail*. 2024;12:1625–1635. doi: 10.1016/j.jchf.2024.06.009
9. Ton VK, Li S, John K, Li B, Zweck E, Kanwar MK, Sinha SS, Hernandez-Montfort J, Garan AR, Goodman R, et al. Serial shock severity assessment within 72 hours after diagnosis: a cardiogenic shock working group report. *J Am Coll Cardiol*. 2024;84:978–990. doi: 10.1016/j.jacc.2024.04.069
10. Zweck E, Ton VK, Kanwar M, Li S, Li B, Sinha SS, Hernandez-Montfort J, Garan AR, Abraham J, Blumer V, et al; CSWG Academic Research Consortium. CSWG-SCAI stages combined with machine learning-based phenotypes for serial risk stratification in cardiogenic shock. *JACC Heart Fail*. 2025;13:102611. doi: 10.1016/j.jchf.2025.102611
11. Varshney AS, Palazzolo MG, Barnett CF, Bohula EA, Burke JA, Chaudhry SP, Chonde MD, Ghafghazi S, Gerber DA, Kenigsberg B, et al. Epidemiology and prognostic significance of acute noncardiac organ dysfunction across cardiogenic shock subtypes. *J Card Fail*. 2025;31:1512–1522.
12. Mehta C, Shin A, Osorio B, DePolo D, Vargas I, Hao E, Khan A, Chandragiri S, Shringi S, McLean Diaz PO, et al. Management of noncardiac organ failure in cardiogenic shock. *Am Heart J Plus*. 2025;55:100549. doi: 10.1016/j.ahjo.2025.100549
13. Jentzer JC, Burstein B, Van Diepen S, Murphy J, Holmes DR Jr, Bell MR, Barsness GW, Henry TD, Menon V, Rihal CS, et al. Defining shock and preshock for mortality risk stratification in cardiac intensive care unit patients. *Circ Heart Fail*. 2021;14:e007678. doi: 10.1161/CIRCHEARTFAILURE.120.007678
14. Kapur NK, Kanwar M, Sinha SS, Thayer KL, Garan AR, Hernandez-Montfort J, Zhang Y, Li B, Baca P, Dieng F, et al. Criteria for defining stages of cardiogenic shock severity. *J Am Coll Cardiol*. 2022;80:185–198. doi: 10.1016/j.jacc.2022.04.049
15. Lameire NH, Levin A, Kellum JA, Cheung M, Jadoul M, Winkelmayer WC, Stevens PE; Conference Participants. Harmonizing acute and chronic kidney disease definition and classification: report of a kidney disease: improving global outcomes (KDIGO) consensus conference. *Kidney Int*. 2021;100:516–526. doi: 10.1016/j.kint.2021.06.028
16. Vallabhajosyula S. End-organ injury and failure: the true danger in cardiogenic shock. *Circulation*. 2024;150:2004–2006. doi: 10.1161/CIRCULATIONAHA.124.072571
17. Wiens EJ, Jentzer JC, Miller PE, Vallabhajosyula S, Berg DD, Bagshaw SM, van Diepen S. Renal replacement therapy in patients with cardiogenic shock: refining the AEIOUs in cardiovascular patients. *Can J Cardiol*. 2025;41:779–783. doi: 10.1016/j.cjca.2024.11.004
18. Sakhuja A, Bandak G, Barreto EF, Vallabhajosyula S, Jentzer J, Albright R, Kashani KB. Role of loop diuretic challenge in stage 3 acute kidney injury. *Mayo Clin Proc*. 2019;94:1509–1515. doi: 10.1016/j.mayocp.2019.01.040
19. Kochar A, Vallabhajosyula S, John K, Sinha SS, Esposito M, Pahuja M, Hirst C, Li S, Kong Q, Li B, et al. Factors associated with acute limb ischemia in cardiogenic shock and downstream clinical outcomes: insights from the Cardiogenic Shock Working Group. *J Heart Lung Transplant*. 2024;43:1846–1856.
20. Kataria R, Sinha SS, Li S, Kong Q, Kanwar M, Hernandez-Montfort J, Garan AR, Abraham J, Zweck E, Ton VK, et al. Worsening renal function is common and is associated with higher mortality rates in cardiogenic shock: a Cardiogenic Shock Working Group Report. *J Card Fail*. 2025; doi: 10.1016/j.cardfail.2025.03.012
21. Vallabhajosyula S, Shankar A, Patlolla SH, Prasad A, Bell MR, Jentzer JC, Arora S, Vallabhajosyula S, Gersh BJ, Jaffe AS, et al. Pulmonary artery catheter use in acute myocardial infarction-cardiogenic shock. *ESC Heart Fail*. 2020;7:1234–1245. doi: 10.1002/ehf2.12652
22. Hernandez-Montfort J, Kanwar M, Sinha SS, Garan AR, Blumer V, Kataria R, Whitehead EH, Yin M, Li B, Zhang Y, et al. Clinical presentation

and in-hospital trajectory of heart failure and cardiogenic shock. *JACC Heart Fail*. 2023;11:176–187. doi: 10.1016/j.jchf.2022.10.002
23. Harrison AM, Thongprayoon C, Kashyap R, Chute CG, Gajic O, Pickering BW, Herasevich V. Developing the surveillance algorithm for detection of failure to recognize and treat severe sepsis. *Mayo Clin Proc*. 2015;90:166–175. doi: 10.1016/j.mayocp.2014.11.014
24. Papolos AI, Kenigsberg BB, Berg DD, Alviar CL, Bohula E, Burke JA, Carnicelli AP, Chaudhry SP, Drakos S, Gerber DA, et al; Critical Care Cardiology Trials Network Investigators. Management and outcomes of cardiogenic shock in cardiac ICUs with versus without shock teams. *J Am Coll Cardiol*. 2021;78:1309–1317. doi: 10.1016/j.jacc.2021.07.044
25. Vallabhajosyula SMC, Jindal A, Mehta A, Khetpal V, Jentzer JC, Ton VK, Patel PC, Naidu SS, Kapur NK, Abbott JD. Cardiogenic shock detection using electronic medical records: a review and blueprint for clinical implementation and future research. *J Am Heart Assoc*. 2025. In Press
26. Hill KL, Rustin MA, Asche MA, Bennett CE, Patel PC, Jentzer JC. Cardiogenic shock classification and associated mortality risk. *Mayo Clin Proc*. 2023;98:771–783. doi: 10.1016/j.mayocp.2022.12.007
27. Vallabhajosyula S, Dunlay SM, Barsness GW, Vallabhajosyula S, Vallabhajosyula S, Sundaragiri PR, Gersh BJ, Jaffe AS, Kashani K. Temporal trends, predictors, and outcomes of acute kidney injury and hemodialysis use in acute myocardial infarction-related cardiogenic shock. *PLoS One*. 2019;14:e0222894. doi: 10.1371/journal.pone.0222894
28. Vallabhajosyula S, Ya'Qoub L, Dunlay SM, Vallabhajosyula S, Vallabhajosyula S, Sundaragiri PR, Jaffe AS, Gersh BJ, Kashani K. Sex disparities in acute kidney injury complicating acute myocardial infarction with cardiogenic shock. *ESC Heart Fail*. 2019;6:874–877. doi: 10.1002/ehf2.12482
29. Jentzer JC, Bihorac A, Brusca SB, Del Rio-Pertuz G, Kashani K, Kazory A, Kellum JA, Mao M, Moriyama B, Morrow DA, et al; Critical Care Cardiology Working Group of the Heart Failure and Transplant Section Leadership Council. Contemporary management of severe acute kidney injury and refractory cardiorenal syndrome: JACC council perspectives. *J Am Coll Cardiol*. 2020;76:1084–1101. doi: 10.1016/j.jacc.2020.06.070
30. Singh S, Kanwar A, Sundaragiri PR, Cheungpasitporn W, Truesdell AG, Rab ST, Singh M, Vallabhajosyula S. Acute kidney injury in cardiogenic shock: an updated narrative review. *J Cardiovasc Dev Dis*. 2021;8:88. doi: 10.3390/jcdd8080088
31. Naidu SS, Nathan S, Basir MB, Baran DA, Marbach JA, Grines CL. SCAI door to lactate clearance (SCAI DLC) cardiogenic shock initiative: definition, hypothesis, and call to action. *J Soc Cardiovasc Angiogr Interv*. 2025;4:103996. doi: 10.1016/j.jscai.2025.103996

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